

School of Chemical Engineering

CHEE3007- Process Modelling & Dynamics

Semester 2, 2023 Practical

Briefing Sheet



THE UNIVERSITY OF QUEENSLAND
A U S T R A L I A

1. Learning Objectives

Welcome to CHEE3007 Process Modelling and Dynamics Practical. The overall objective of this practical is to investigate the impact of input variables on output variables in a distillation process. By conducting experiments and analysing the data collected, you will gain an understanding of how changes in input variables affect the overall performance and efficiency of the distillation process. Furthermore, you will learn how to utilise the data obtained as input for developing dynamic model and implementing control strategies in your model-building assignments (Project 1 and Project 2). Through this practical experience, you will enhance your knowledge of distillation processes, data analysis, and control implementation, enabling you to apply these skills in real-world scenarios.

1.1 Assessment Component

After each practical you must complete and submit the Practical Submission Sheet answering questions by using experiences collected from the laboratory practical. The empty submission sheet can be found on blackboard.

1.2 Submission

Write your response under each question and submit it to the tutors upon completion of the practical. The submission must be made at the end of the practical session. The practical component is worth a total of 1.5 marks.

1.3 Support

You can request support or post questions about the practical on the Ed Discussion Board on Blackboard.

2. Practical Briefing

We shall familiarize ourselves with the Pignat Pilot Distillation Column situated at 46-169, as illustrated in Figure 1. Prior to attending the practical you must have completed the online learning module:

- UQ undergraduate lab induction

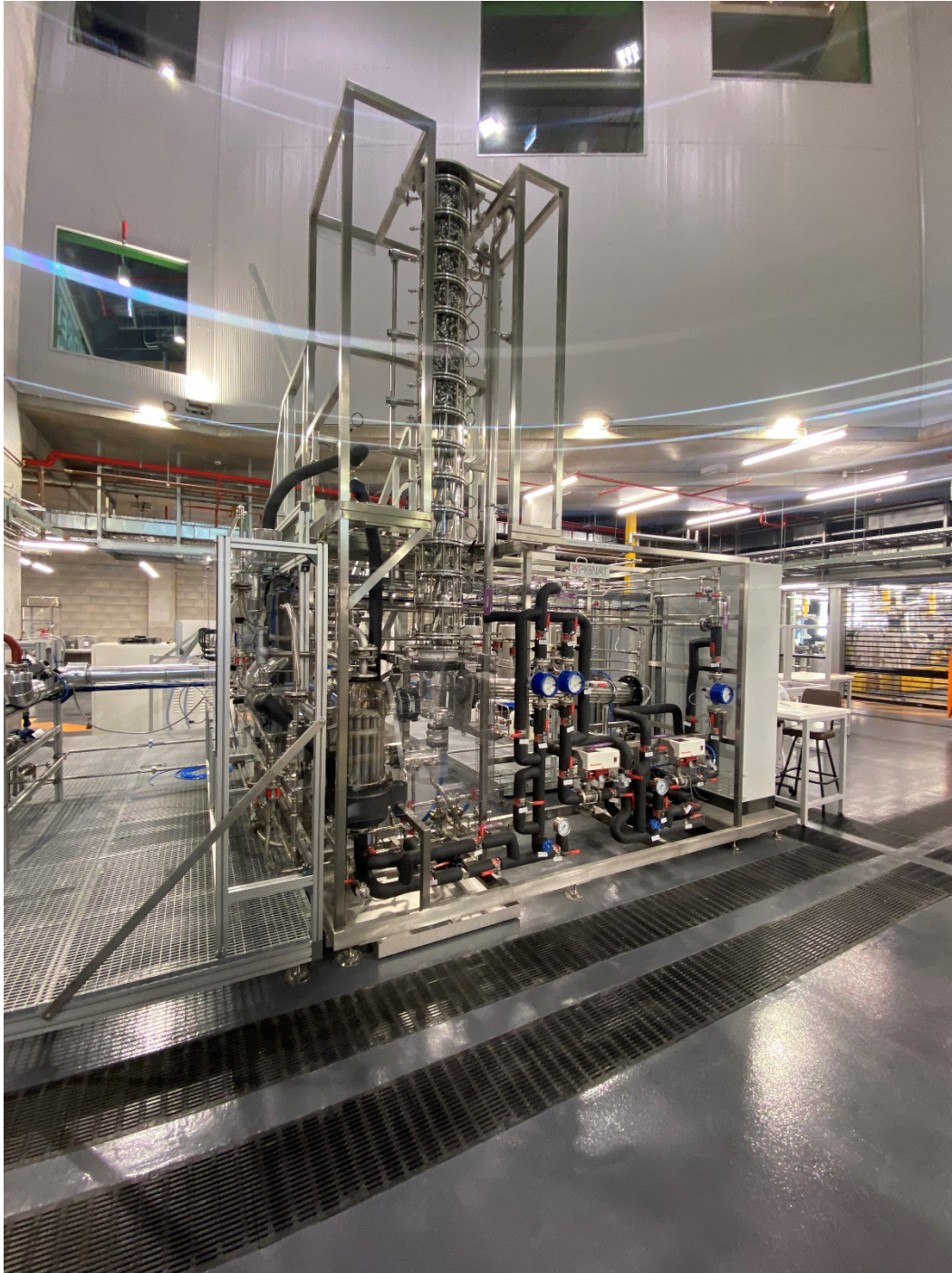


Figure 1 PIGNAT Distillation Column

Resources Required

- PIGNAT distillation column rig.
- Refractometer
- Mono-ethylene glycol
- Water
- DeltaV monitoring and data collection system.

3. Safety

Please consider numerous risks linked to the equipment (such as electrocution, spillage, and falling, among others as described in risk assessment attached). The utmost importance is given to ensuring lab safety for EVERY student throughout the lab sessions. When present in the lab, it is obligatory to wear the required PPE which include:

1. Trousers, and closed shoes.
2. Lab coat
3. Safety Glasses
4. Gloves

- **Electrical Safety:** All the equipment utilized in this unit operates using a standard 240 V main supply line in close vicinity to water. The main supply is provided with RCD protection. Students are not supposed to handle any main electrical units connecting the instruments. If you come across any unwarranted situation, then immediately alert the lab assistant/instructor.
- **Chemical Safety:** The practical uses hazardous chemicals such as mono-ethylene glycol, but the laboratory does have chemical hazards. In addition, you are not required to personally prepare any of the chemicals used. In consideration of this, all PPE except gloves must be always worn. In addition, gloves must be worn when conducting sampling of the distillate and the bottoms product. They can be removed at table areas when not touching practical equipment. Refer to material safety datasheet (MSDS) of mono-ethylene glycol attached in the Appendix A1.
- **Mechanical Safety:** Ensure that long hair is securely tied back and, if it is particularly long, either tucked into the lab coat or tied up. There should be no lifting involved in the practical activities. If any task requires you to lift objects or position your body in an uncomfortable manner, immediately inform the tutor.
- **Personal and process safety:** Be cautious of hot surfaces (boiler, steam piping, and reboiler) during practicals. Steam pipes have insulating jackets for user protection and heat conservation. The reboiler has a front protection panel to prevent burns. Also watch out for slip, trip, and fall hazards, including stairs. Due to the presence of noise, it is advisable to wear earmuffs or earplugs for hearing protection. Avoid using phones near the equipment to prevent interference and accidents.

Figure 2 shows various safety hazards in the distillation column, such as cautions, trip hazards, and hot surfaces.

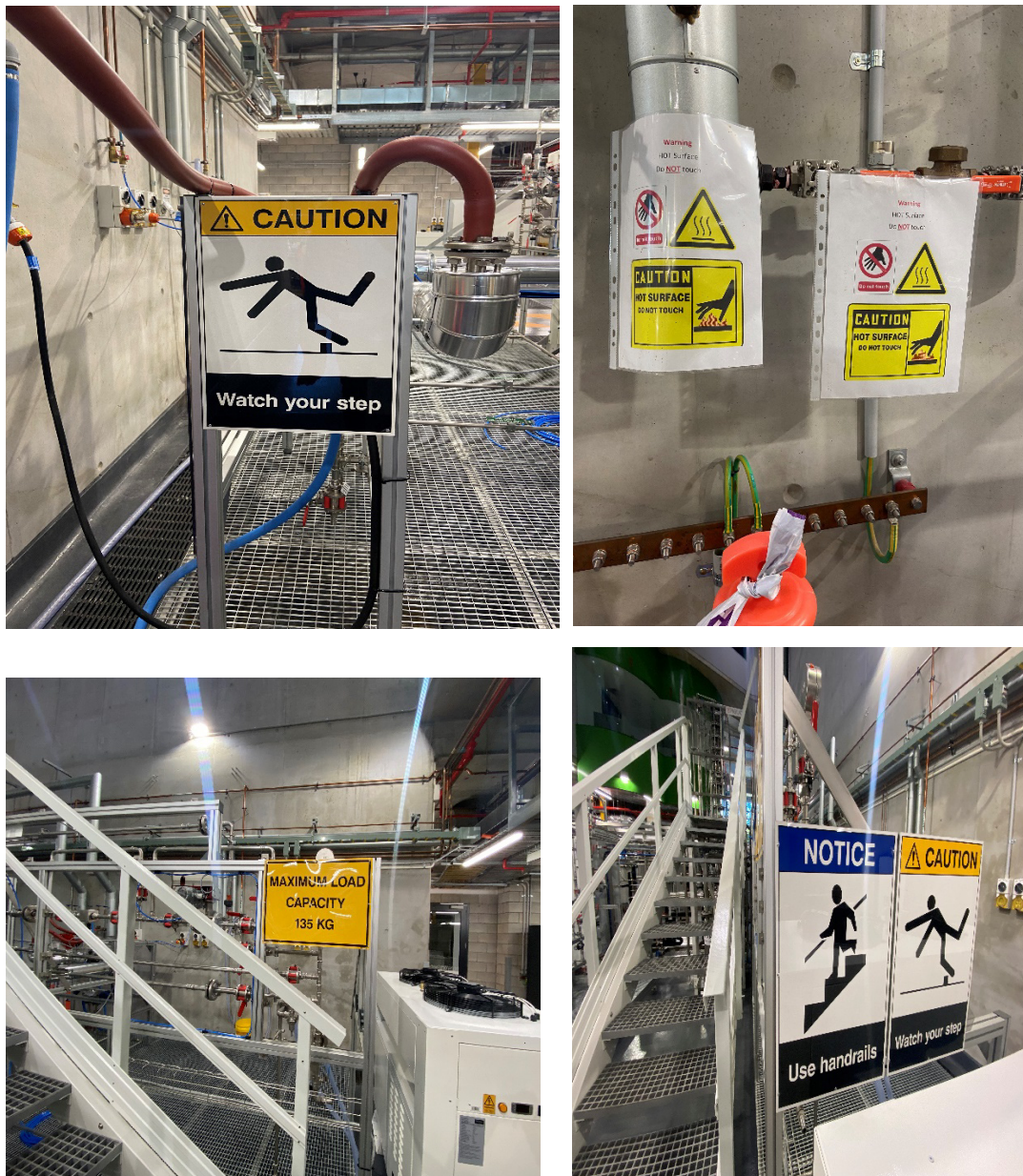


Figure 2. Various safety hazards around the distillation column

4. Background

The undergraduate teaching laboratory in the School of Chemical Engineering at the University of Queensland is equipped with a PIGNAT Pilot Distillation Plant, commonly referred to as a distillation column. This equipment imitates the control and operation of industrial distillation columns. To gain a better understanding of how this column works, refer to the Piping and Instrumentation Diagram (P&ID) in the Operation Manual provided in the Appendix A2.

The distillation column in the laboratory has seven bubble-cap trays that maintain a liquid level on each tray through overflow. The caps force vapours to pass through slots below the liquid level, creating bubbles as they rise. Additionally, there are seven packed trays in the upper section of the column. In contrast, industrial units typically have many caps on each tray or many packed columns.

Unlike most industrial units, the laboratory equipment is equipped with three sampling valves positioned at valve V32 on the bottom product line, valve V58 on the distillate product tank line, and valve V10 on the feed tank line. These sampling points have thermocouples for temperature recording.

The DeltaV monitoring system for the distillation column is located on the control cabinet's right-hand side. Furthermore, the distillation process can be remotely controlled using a computer connected via Ethernet cable or wireless network.

The main components of a distillation column include:

1. **Reboiler:** It is a heat exchanger that transfers heat from a heating fluid, often steam, to the liquid mixture undergoing distillation. This causes a fraction of the liquid in the reboiler to vaporize, rising upward and driving further separation within the distillation column.
2. **Distillation Column:** Also known as a tower, it is a vertical cylindrical structure with multiple layers of column internals. Tray columns and packed columns are the two types of distillation columns based on the design and mechanical nature of the internals. Tray columns have mechanical trays arranged in a cascade, while packed columns have randomly dumped or structured packings. Both types are widely used in various industries to separate liquid and gas mixtures by facilitating gas-liquid contact. Tray columns operate stage-wise, with mass transfer occurring on individual trays. Packed columns involve differential mass transfer, with continuous mass transfer from the bottom to the top of the column between the liquid and vapor streams flowing counter currently.
3. **Condenser:** It is a heat exchanger that converts process vapor back into a liquid state using a cooling medium. In a distillation column, a condenser transfers latent heat from the vapor stream exiting the top of the column to a utility stream like cooling water or air. This converts the liquid condensate into reflux and top product.
4. **Feed Line:** The feed line of a distillation column includes feed storage, flow control, a feed conditioning unit (typically a heat exchanger), and a distribution manifold. These components are vital for distillation operations as the feed flow rate, condition, and location significantly influence the column's operation, performance, and design.
5. **Overhead Reservoir or Drum:** This vessel in the distillation process serves several important roles. It maintains a desired level of overhead liquid to prevent vapor from escaping the condenser, stores overhead product, and its storage level acts as input for overhead flow control. Additionally, it functions as a decanter when the overhead product is a two-phase mixture.
6. **Bottom Product Flow:** The withdrawal of liquid product from the bottom of the reboiler involves devices and instruments. This aspect plays a crucial role in reboiler design and operation.

7. **Utilities:** These are essential components for distillation operation and design, including electricity, high-pressure air for pneumatic valves, and notably, process steam and cooling water.

For more information refer to the piping and instrumentation diagram given in Appendix A2.

5. Experimental Procedure

5.1 Overview

In this practical session, you will run four experiments:

1. Run the distillation column at the initial steady state conditions.
2. Induce system disturbance by sequentially changing the feed flow rate, reflux ratio, feed tray location, and steam pressure.

For each disturbance, measure the designated process parameters as instructed by the experimental recording sheet. Employ a refractometer to measure the glycol concentration in the feed, bottom, and distillate products.

5.2 Experimental Steps

The column will be operating at a steady state before we conduct the experiments defined below.

5.2.1 Experiment I: Changing Feed Flowrate

1. Adjust the feed flow rate using the DeltaV monitoring software by modifying the set point on FLC01.
2. Allow sufficient time for the column temperatures to stabilize (TT01 to TT15).
3. Record all the parameters displayed on the interface, as specified in the provided recording sheets.
4. To identify the sampling points for feed, distillate, and bottoms composition, utilize valves V10, V58, and V32.
5. Collect samples from the designated sampling points and employ a refractometer to determine the ethylene glycol concentration.
6. After collecting the required data bring the system back to original configuration.

5.2.2 Experiment II: Changing Reflux Flowrate

1. Adjust the reflux flow rate using the DeltaV monitoring software by modifying the set point on FLC01.
2. Allow sufficient time for the column temperatures to stabilize (TT01 to TT15).
3. Repeat Step 3 to 6 from Experiment I.

5.2.3 Experiment III: Changing Feed tray entry location.

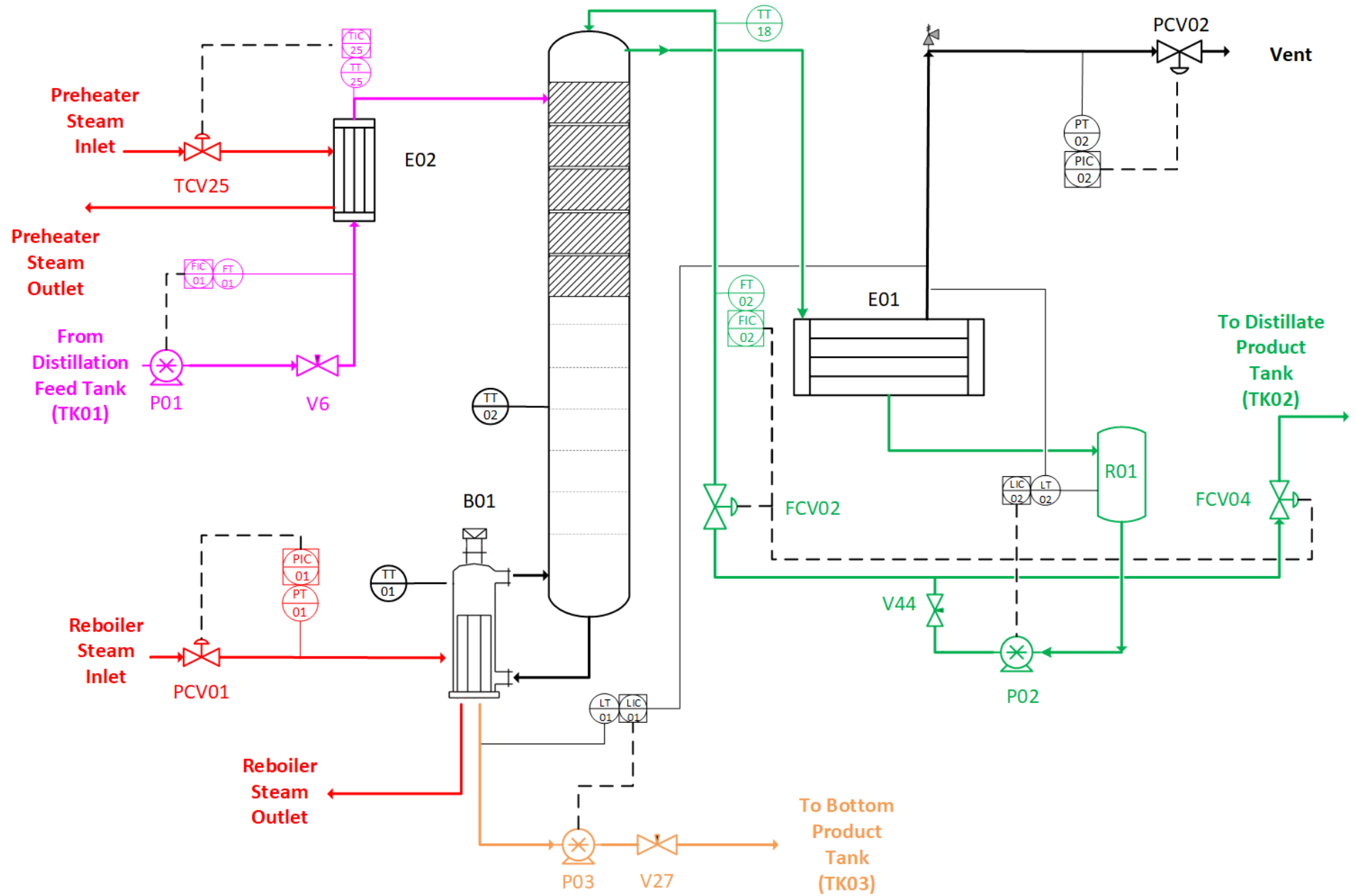
1. Change the feed tray location by physically adjusting the inlet valve from V16 to V14. Open V14 and close V16 in succession.
2. Allow sufficient time for the column temperatures to stabilize (TT01 to TT15).
3. Repeat Step 3 to 6 from Experiment I.

5.2.4 Experiment IV: Changing Steam pressure.

1. Adjust the steam pressure using the DeltaV monitoring software by modifying the set point on PIC01.
2. Allow sufficient time for the column temperatures to stabilize (TT01 to TT15).
3. Repeat Step 3 to 6 from Experiment I.

<i>Record these values once the column has reached a steady state:</i>	Current State	Effect of Reflux Ratio	Effect of Inlet Flow rate	Effect of Feed Position/Tray Number	Effect of Steam Pressure
Recording Time: HOUR:MIN					
Column temperatures, deg C; T1 to T15					
Feed inlet Temperature, deg C					
Feed Flow Rate, L/h					
Distillate Flow Rate, L/h					
Bottoms Flow Rate, L/h					
Reflux Flow Rate, L/h					
Reflux Ratio					
Feed composition, wt.% (H ₂ O:EG)					
Distillate Composition, wt. % (H ₂ O)					
Bottom Product composition wt.% (EG)					
Steam Flow rate, kg/h					
Steam temperature, deg C Inlet and Outlet					
Steam Pressure, bar					
Column Pressure, bar					
Column differential Pressure, bar					
Condenser Volume, L					
Reboiler volume, L					
Feed Position, tray number					

A2. Piping and Instrumentation Diagram of Distillation Column



Appendix

A1. MSDS Ethylene glycol



ETHYLENE GLYCOL

ChemWatch Review SDS

Chemwatch: 10084
Version No: 4.1
Safety Data Sheet according to WHS Regulations (Hazardous Chemicals) Amendment 2020 and ADG requirements

Chemwatch Hazard Alert Code: 3

Issue Date: 23/12/2022
Print Date: 27/06/2023
L.GHS.AUS.EN.RISK

SECTION 1 Identification of the substance / mixture and of the company / undertaking

Product Identifier

Product name	ETHYLENE GLYCOL
Chemical Name	ethylene glycol
Synonyms	CH ₂ -OHCH ₂ -OH; C ₂ H ₆ O ₂ ; ethylene glycol antifreeze; ethanediol; 1,2-ethanediol; UCAR 17; ethandiol; 1,2-dihydroxyethane; ethylene dihydrate; glycol oglycol alcohol; Lustrol-9; M.E.G.; monoethylene glycol; Norkool; Tescol; Dowtherm SR1; Merck ethanediol AnalaR; ethylene glycol; Thermal Arc Torch Coolant Concentrate
Chemical formula	C ₂ H ₆ O ₂ [C ₂ H ₆ O ₂
Other means of identification	Not Available
CAS number	107-21-1

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses	Used as a coolant and antifreeze in heating and cooling systems; heat transfer agent; as an industrial humectant; as a solvent in the paint and plastics industries, leather dyeing and textile processing and for preparation of polyesters. Also used in the formulation of printers' inks and ink for ball-point pens; in the synthesis of safety explosives and polyester fibres. [~Intermediate ~]
--------------------------	---

Details of the manufacturer or supplier of the safety data sheet

Registered company name	Redox	Special Finishes
Address	3960 Paramount Boulevard, Suite 107 Lakewood CA 90712 United States	16 Culperry Road Glendene, Waitakere Auckland 0602 New Zealand
Telephone	+1 424 675 3200	+64 9 818 9595
Fax	+1 424 675 3211	+64 9 818 9892
Website	http://redox.com/	http://specialfinishes.co.nz
Email	info@redox.com	sales@southsidecleaning.com.au

Emergency telephone number

Association / Organisation	Redox	CHEMWATCH EMERGENCY RESPONSE (24/7)
Emergency telephone numbers	+1 201 353 3365	+61 1800 951 288
Other emergency telephone numbers	Not Available	+61 3 9573 3188

Once connected and if the message is not in your preferred language then please dial 01

SECTION 2 Hazards identification

Classification of the substance or mixture

HAZARDOUS CHEMICAL. NON-DANGEROUS GOODS. According to the WHS Regulations and the ADG Code.

Chemwatch Hazard Ratings

	Min	Max
Flammability	1	Blue
Toxicity	2	Blue
Body Contact	3	Blue
Reactivity	1	Blue
Chronic	2	Blue

NFPA 704 diamond



Note: The hazard category numbers found in GHS classification in section 2 of this SDSs are NOT to be used to fill in the NFPA 704 diamond. Blue = Health Red = Fire Yellow = Reactivity White = Special (Oxidizer or water reactive substances)

Poisons Schedule	S6
Classification [1]	Acute Toxicity (Oral) Category 4, Serious Eye Damage/Eye Irritation Category 2B, Specific Target Organ Toxicity - Repeated Exposure Category 2 *LIMITED EVIDENCE
Legend:	1. Classified by Chemwatch; 2. Classification drawn from HCIS; 3. Classification drawn from Regulation (EU) No 1272/2008 - Annex VI

Chemwatch: 10084
Version No: 4.1

Page 2 of 13

ETHYLENE GLYCOL

Issue Date: 23/12/2022
Print Date: 27/06/2023

Label elements

Hazard pictogram(s)	
Signal word	Warning

Hazard statement(s)

H302	Harmful if swallowed.
H320	Causes eye irritation.
H373	May cause damage to organs through prolonged or repeated exposure.

*LIMITED EVIDENCE

Supplementary statement(s)

Not Applicable

Precautionary statement(s) Prevention

P260	Do not breathe mist/vapours/spray.
P264	Wash all exposed external body areas thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.

Precautionary statement(s) Response

P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P314	Get medical advice/attention if you feel unwell.
P337+P313	If eye irritation persists: Get medical advice/attention.
P301+P312	IF SWALLOWED: Call a POISON CENTER/doctor/physician/first aider if you feel unwell.
P330	Rinse mouth.

Precautionary statement(s) Storage

Not Applicable

Precautionary statement(s) Disposal

P501	Dispose of contents/container to authorised hazardous or special waste collection point in accordance with any local regulation.
------	--

SECTION 3 Composition / information on ingredients

Substances

CAS No	%[weight]	Name
107-21-1	>99	ethylene glycol

Legend: 1. Classified by Chemwatch; 2. Classification drawn from HCIS; 3. Classification drawn from Regulation (EU) No 1272/2008 - Annex VI; 4. Classification drawn from C&L;
* EU IOELVs available

Mixtures

See section above for composition of Substances

SECTION 4 First aid measures

Description of first aid measures

Eye Contact	If this product comes in contact with the eyes: - Immediately hold eyelids apart and flush the eye continuously with running water. - Ensure complete irrigation of the eye by keeping eyelids apart and away from eye and moving the eyelids by occasionally lifting the upper and lower lids. - Continue flushing until advised to stop by the Poisons Information Centre or a doctor, or for at least 15 minutes. - Transport to hospital or doctor without delay. - Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.
Skin Contact	If skin or hair contact occurs: - Immediately flush body and clothes with large amounts of water, using safety shower if available. - Quickly remove all contaminated clothing, including footwear. - Wash skin and hair with running water. Continue flushing with water until advised to stop by the Poisons Information Centre. - Transport to hospital, or doctor.
Inhalation	- If fumes or combustion products are inhaled remove from contaminated area. - Lay patient down. Keep warm and rested. - Prostheses such as false teeth, which may block airway, should be removed, where possible, prior to initiating first aid procedures. - Apply artificial respiration if not breathing, preferably with a demand valve resuscitator, bag-valve mask device, or pocket mask as trained. Perform CPR if necessary. - Transport to hospital, or doctor, without delay.

Continued...